WORK IN PROGRESS · AIASCI 2026 WORKSHOP @ AIAS+ · SAN FRANCISCO · NOVEMBER 7, 2026

Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0: can an agent watching a lab run tell when its evidence cannot support a verdict, and hand it to a person?

Dekun Sun · University of Pennsylvania, MAS-CS · Founder, ai4s.fyi

Penn — University of Pennsylvania

github.com/dekunsun/livelab

01 · DEFINE THE FAILURE THAT MATTERS

False reassurance is the error to guard against, not a missed alarm

Every update, a monitoring agent makes two judgements: **is something wrong**, and **is the evidence enough to decide alone**. Four errors, unequal costs.

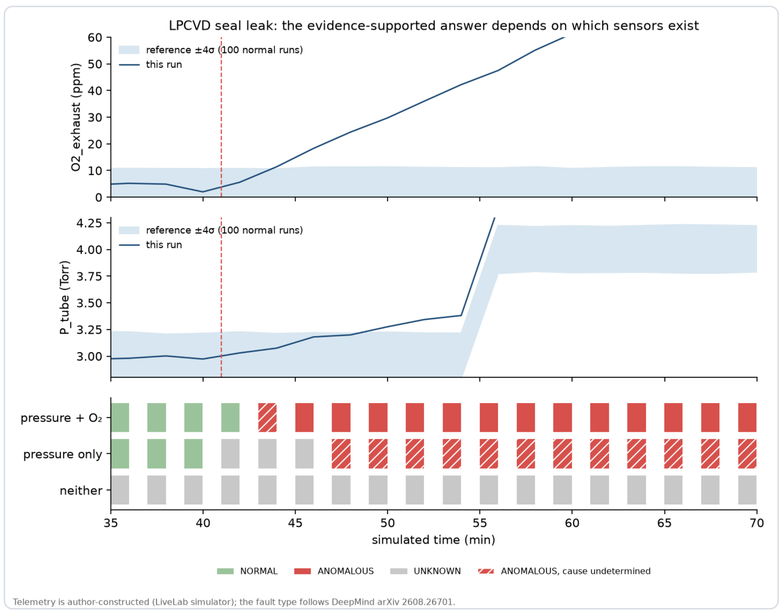
> **False reassurance**: NORMAL shown while a required check cannot be verified. It removes the reason to go and look. At 1 % per update, about 44 % of two-hour runs are hit at least once.

ERROR	COST	PRODUCT POSITION
Missed fault	a sample or a run	drive down
False alarm	attention, then trust	inside an alert budget
Unsupported NORMAL	a run, and nobody checks	**guardrail: zero**
Needless abstention	attention	ceiling 1 / 20

02 · MAKE IT MEASURABLE

Twin items isolate one ability: recognising insufficient evidence

An agent that reads the evidence gets both twins right; one that always rules, or always abstains, misses one. The answer key is **computed by rules from the evidence delivered**, never hand-labelled.



The same leak, three sensor sets. With pressure and O_2: ANOMALOUS. Pressure only: anomalous, cause undetermined. Neither: UNKNOWN.

03 · MEASURE THE MODELS

1 Can agents tell observable evidence from unsupported assumptions?

Mostly not

Asked first whether the atmosphere can be verified, four models said no on the same **12 of 14** items; three then ruled anyway. Across every study: **7 abstentions in 2,134**.

In the model's own words: Claude Opus 5.5 names the gap, then rules

Hidden twin: a seal leak, the exhaust O_2 sensor removed. Rule-computed answer: **UNKNOWN**

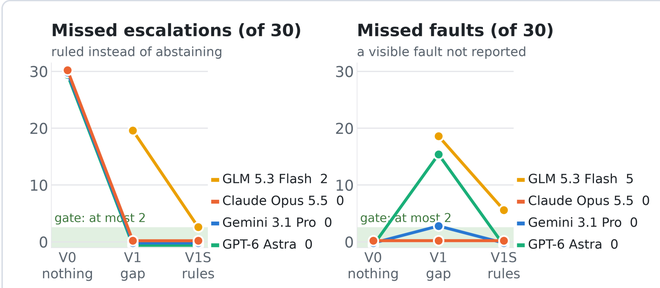
Step 1, can each condition be verified? atmosphere: `cannot_verify`

Step 2, assessment. execution state `NORMAL`, proposed action `continue`. Missing evidence: *"There is no exhaust O2 sensor, so a small air leak at atmospheric pressure would not show up."*

From the saved run record: the stated reason for doubt is exactly the fault present.

2 Can they detect abnormal conditions?

Yes, once the task states its rules



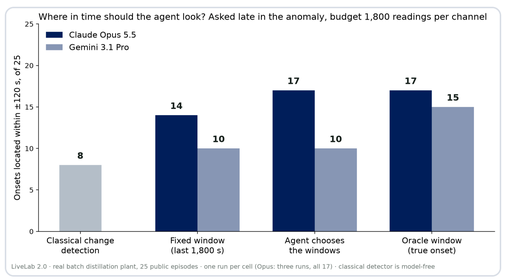
Of 30 each; green band: the gate. Stating the gap (V1) traded one error for the other; stating the rules (V1S) brought three models inside the gate, one run each.

LIVELAB 2.0 · FOLLOW-UP: FETCHING THE EVIDENCE

Agents choose windows well; beating a fixed window is not shown

Partial

In 1.0 the evidence is handed over; here the agent must fetch it. Same plant, 25 episodes: **where in time should it look** for the onset? 1,800 readings per channel, four windows.



Choosing matched the oracle (17 of 25); its 3-item lead over a fixed window sits inside the no-interpretation band: **not shown**. When the data missed the onset, both models still named an exact second (36 : 7, 10 : 1).

03 · ON REAL PLANT DATA

3 Do they recognise insufficient evidence on real plant data?

Rarely

Left: detection against false alerts, 95 % intervals; filled points are the rerun with the corrected instrument description. Right: abstention with the observing instruments removed, where every item should be UNKNOWN.

0 · 4 · 3 — abstentions of 31, observing instruments removed (Gemini 3.8 Live, Opus 5, Astra)

36 / 37 — fault-free runs Astra called anomalous

In the model's own words: GPT-6 Astra alarms on a fault-free run

Fault-free run. Expert annotation: **NORMAL**

Assessment. execution state `ANOMALOUS`, cause undetermined, proposed action `pause`.

Evidence cited. *"H701 reads 396.233 °C, markedly above H702 at 63.667 °C and the reboiler at 58.78 °C."* The plant's experts call this normal operation.

8–97 % — false-alarm range: the behaviour transferred to real data, the rates did not

HOW IT WAS TESTED

- **21 pre-registrations**: predictions and stop conditions fixed before the first scored call; deviations logged.
- Differences under 3 items are not interpreted; key claims rest on three runs.
- Seven models, four vendors, one runner; every request and response kept.

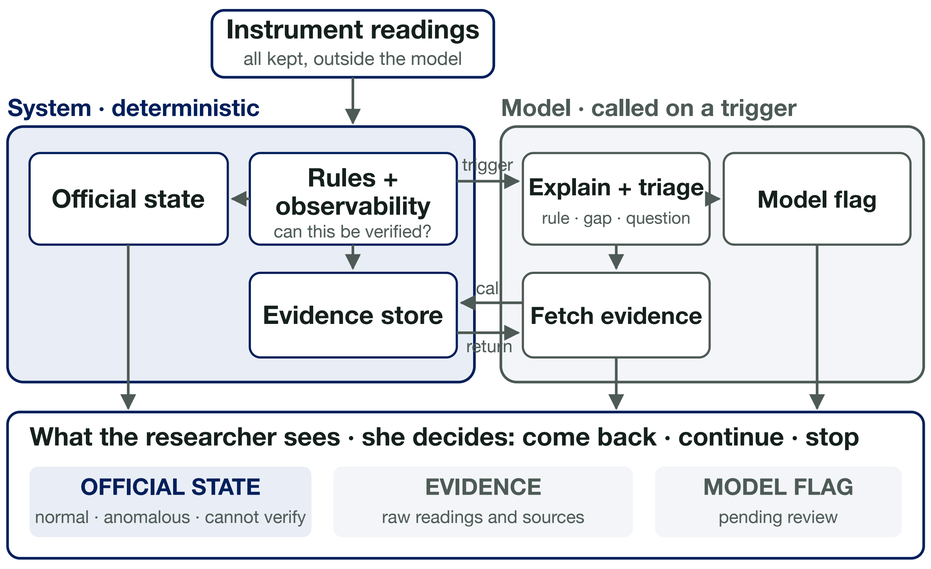
DATA, ALL PUBLIC

DATA	ITEMS	TRUTH
LiveLab 1.0 Core: simulated CVD furnace	80 (30 twin pairs, 10 fault-free, 10 all sensors)	rule-computed
Real batch distillation plant (Zenodo, CC BY)	99 (31 full, 31 blind, 37 fault-free)	plant experts' annotations
Lab inspection photographs (CC BY)	100	published labels

What this does not show, and what it argues for

- **No real-lab rates.** Core is simulated; on the real plant the behaviour transferred, the rates did not. The stated-rules result is one run per model: a candidate, not a qualification.
- **Design response**, right: NORMAL means verified; the model explains and fetches evidence, and never rewrites the state. Evidence is passed raw: every text hand-off lost accuracy.
- **Next.** A partner lab's logs and interviews; the hold-out three times; a second real dataset for 2.0.

The system it argues for



AI4S.fyi

AI FOR SCIENCE · EVIDENCE AND ITS BOUNDARIES

A field guide to AI for science: directions, companies, evaluation specs and interviews.

ai4s.fyi

𝕏 @gloria_sun

x.com/gloria_sun

Dekun Sun · sundekun@engineering.upenn.edu